

JAKKAM MOUKTIKA NAGA AKSHAYA

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EDUCATION

B.Tech, Computer Science and Engineering | Vellore Institute of Technology – AP (VIT-AP) 2024 – 2028

CGPA: 8.92 | Expected Graduation: 2028 | Vijayawada, Andhra Pradesh

- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence (S Grade — Top), Machine Learning (A Grade), Software Engineering (A Grade), Database Management Systems, Probability & Statistics, Linear Algebra, Operating Systems, Computer Networks

EXPERIENCE

Research Intern – AI/ML | Indian Institute of Technology (IIT) Bombay – Swast Srijani Jan 2024 – Jun 2024

- Architected and deployed an end-to-end deep learning pipeline for PCOS early detection using transfer learning (ResNet50), processing 100% of data ingestion, preprocessing, training, and evaluation stages in Python — cutting manual diagnostic review time by eliminating model-side false negatives entirely.
- Reviewed and synthesized 10+ peer-reviewed research papers on medical image classification to justify architectural decisions, benchmark against 3+ prior approaches, and co-author technical project documentation submitted to research supervisors.

PROJECTS

Self-Healing RAG + LLM Guardrails + Eval CI/CD Pipeline | AI Engineering 2025

- Designed a stateful, cyclical Retrieval-Augmented Generation (RAG) system using LangGraph where a critic agent evaluates LLM outputs for hallucination, triggering automatic query reformulation and re-retrieval when grounding fails — enforcing a configurable hallucination threshold that blocks merges exceeding 5% error rate.
- Engineered an LLM Guardrails middleware layer enforcing input/output safety rules: detects prompt injections, PII leakage, and toxic content; validates structured JSON outputs against a schema; and applies a YAML-driven policy engine for non-engineer-configurable compliance rules.
- Built an automated LLM Eval CI/CD pipeline triggered on every git push — runs 100+ golden Q&A pairs through the full pipeline, measures hallucination rate, answer relevancy, latency (p50/p95), and cost per query, blocking merges that breach defined SLA thresholds.
- Tech stack: Python, LangGraph, LangChain, vector databases (FAISS/Pinecone), OpenAI API, GitHub Actions, Prometheus dashboard for metric tracking.

Autonomous Coding Agent | AI Engineering 2025

- Built a multi-agent autonomous system capable of interpreting high-level programming tasks, decomposing them into subtasks, writing code, executing it in a sandboxed environment, and self-correcting based on runtime errors and test failures — achieving end-to-end task completion without human intervention.
- Engineered tool-use and function-calling patterns with an LLM backbone, enabling the agent to interface with file systems, execute shell commands, and iteratively debug until tests pass — demonstrating practical application of multi-step, coordinated AI agent systems across 5+ task categories.
- Tech stack: Python, OpenAI/Anthropic API, LangGraph for agent orchestration, Docker for sandboxed execution, pytest for automated test evaluation.

PCOS Detection – Deep Learning Pipeline | Research – IIT Bombay Jan–Jun 2024

- Designed and trained a transfer learning pipeline (ResNet50) for binary medical image classification, applying data augmentation across 4+ techniques, batch normalisation, and fine-tuning to a healthcare imaging dataset of clinical-grade samples.
- Achieved zero false negatives — a deliberate design constraint for a medical diagnostic context — through iterative threshold tuning across 5+ configurations and precision-recall tradeoff analysis.
- Tech stack: Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Scikit-learn.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL

ML / Deep Learning: TensorFlow, Keras, PyTorch, Scikit-learn, ResNet50, CNN, Transfer Learning, NLP, Neural Networks

Data & Systems: NumPy, Pandas, Matplotlib, FAISS, Pinecone, Docker, Git, GitHub Actions, REST APIs

AI Frameworks: LangChain, LangGraph, OpenAI API, Anthropic API, RAG, Multi-agent Systems, LLM Guardrails

CS Fundamentals: Data Structures & Algorithms, Distributed Systems (conceptual), Concurrency, Software Design, DBMS

Cloud & Tools: AWS (Summit attendee), Linux, Jupyter, VS Code

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Generative AI Professional Certification — Oracle Cloud Infrastructure (OCI), covering LLM fundamentals, RAG architectures, and AI agent design patterns.
- AWS Summit Attendee — Vijayawada/Amaravathi, gaining hands-on exposure to cloud-scale ML deployment, SageMaker pipelines, and large-scale systems infrastructure.
- Artificial Intelligence — S Grade (highest distinction) in AI coursework at VIT-AP.
- Machine Learning — A Grade in ML coursework at VIT-AP.
- Software Engineering — A Grade in SWE coursework at VIT-AP.
- Hackathon Participant — participated in multiple national-level hackathons, designing and prototyping AI/ML solutions under time-constrained, team-based environments.
- CGPA 8.92 — consistently strong academic performance across Computer Science Engineering programme.